



## CS6660: MATHEMATICAL FOUNDATIONS OF DATA SCIENCE (PROBABILITY)

QUIZ 02

DATE: 26 SEPTEMBER 2026

1. (3 Marks)

Let  $X$  and  $Y$  be jointly uniformly distributed over the triangular region with vertices  $(0, 0)$ ,  $(1, 0)$ , and  $(0, 2)$ . Compute the value of

$$\mathbb{P}(\{X > 0.3\} \mid \{Y \geq 1\}).$$



2. (2 Marks)

Let  $X$  be a random variable with CDF  $F_X$  as shown in the figure below.

Compute the value of

$$\mathbb{P}(\{X \leq 10\} \mid \{X > 2\}).$$

